



Copyright Project Presentation

On

ONLINE EXAMINATION AND EVALUATION SYSTEM

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PROBLEM STATEMENT



Traditional examination systems are inflexible, resource-intensive, and lack comprehensive analytics. They require significant manual effort for question creation, evaluation, and result compilation. Online examinations often suffer from security vulnerabilities like screen sharing and unauthorized assistance. Most platforms provide limited performance insights beyond basic scores. There's a clear need for a unified, secure, and analytical examination platform that streamlines the entire examination process while offering actionable insights for improvement. Exam Matrix addresses these challenges with robust security features, multi-format question support, and detailed performance analytics for both administrators and students.

PROJECT OVERVIEW



Exam Matrix is a web-based online examination and evaluation system designed to revolutionize how educational institutions conduct assessments. The platform offers role-based access with distinct dashboards for administrators and students, supporting multiple question formats including multiple-choice, text-based, checkbox, and coding questions. It implements robust security measures such as single attempt policies to prevent cheating, while providing detailed performance analytics beyond basic scoring. These analytics offer insights into strengths, weaknesses, and areas for improvement, reducing manual effort through automation and enhancing the assessment experience with actionable data and flexible testing options.

Exam Matrix

Create AccountLog In

Professional Certification

Your All-in-One Exam Platform

Take practice exams, get instant scores, and track your progress with our intuitive dashboard

Most Popular

Practice Exams

Access hundreds of practice questions with detailed explanations

Instant Results

Get immediate feedback and performance analysis

Progress Tracking

A large image showing a graduation ceremony with many caps being thrown in the air in front of a large building.

98%
Pass Rate

10K+
Students

Frontend

- **React.js [1]** : For building dynamic and responsive user interfaces
- **CSS [2]** : For styling and layout design
- **Tanstack React Query [3]**: For efficient API call management and caching

Backend

- **Node.js [4]**: JavaScript runtime for server-side operations
- **Express.js [5]**: Web application framework for building robust APIs
- **Prisma ORM [6]**: Database interaction layer for type-safe queries
- **PostgreSQL [7]**: Relational database for structured data storage

Authentication

- **JWT (JSON Web Tokens) [8]**: Secure authentication and session management

Database

- **Neon dB [9]**: Cloud-native PostgreSQL database service for scalable data storage and management

- **Role-Based Access Control**

Implementing distinct permission levels and customized interfaces for administrators and students to ensure appropriate access to system functionalities while maintaining data security and privacy.

- **Class Management System**

Enabling educators to create virtual classes, generate unique enrollment codes, and manage student membership through approval workflows that streamline the administrative aspects of examination management.

- **MCQ-Based Assessment**

Facilitating efficient evaluation through multiple-choice questions with customizable options, automated scoring, and immediate feedback mechanisms that enhance both teaching and learning processes.

- **Flexible Test Scheduling**

Allowing administrators to schedule examinations, establish deadlines, and configure publication settings while maintaining control over result disclosure and accommodating various academic calendars.

- **Advanced Security Measures**

Implementing multiple layers of exam integrity protection, including blocking multiple browser tabs, disabling copy-paste functionality, and requiring camera and microphone activation during assessments.

KEY FEATURES

- **Secure Testing Environment**

Employing single attempt policies, and other security measures to maintain exam integrity, prevent unauthorized assistance, and ensure fair assessment practices across all examination sessions.

- **Comprehensive Analytics Dashboard**

Providing detailed performance insights, test statistics, and personalized improvement recommendations for both educators and learners through interactive visualizations that highlight trends and patterns.

- **Customizable Profile Management**

Enabling users to maintain and update their personal information while integrating with system permissions and access controls, ensuring personalized experiences and accurate record-keeping throughout the platform.

- **Results Management System**

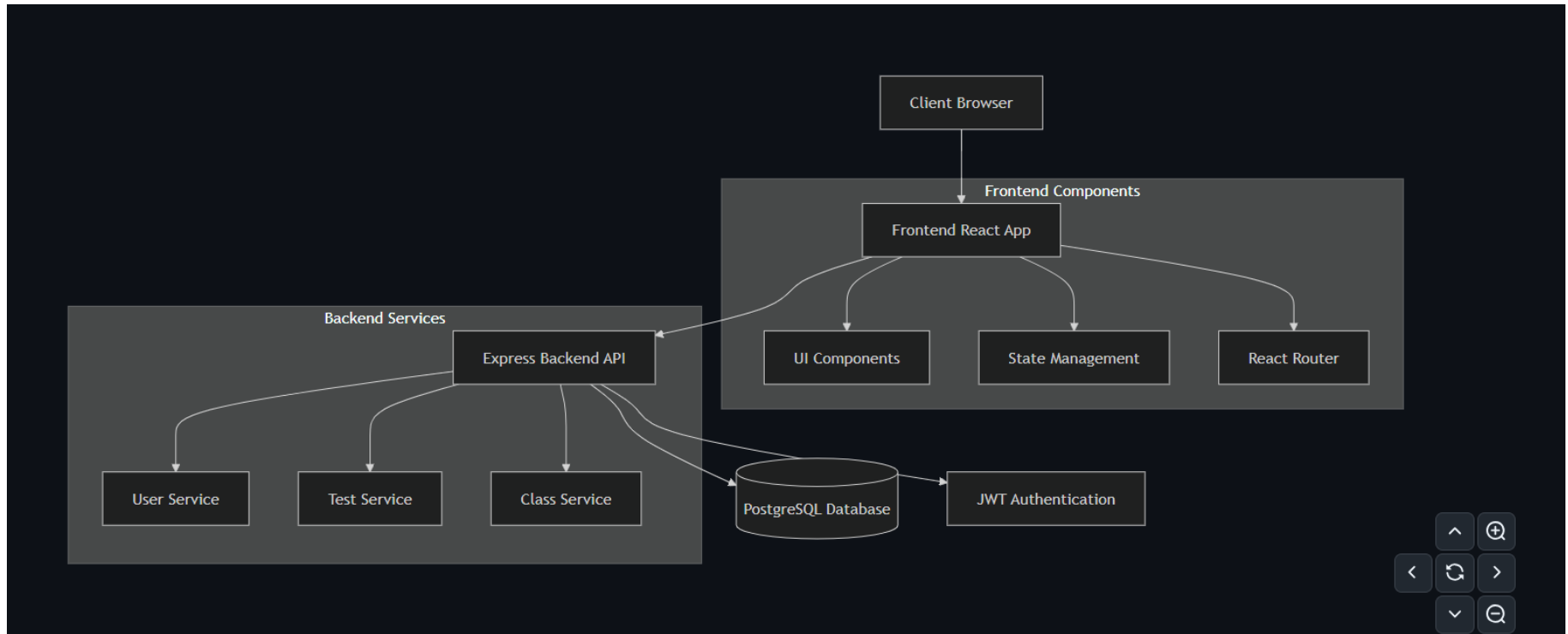
Streamlining the generation, publication, and visualization of examination outcomes through personalized dashboards that display performance metrics, achievements, and historical data to support continuous improvement.

- **Interactive Calendar Interface**

Developing an intuitive calendar dashboard where students and administrators can view upcoming examination schedules, deadlines, and important academic dates with notifications and reminders.

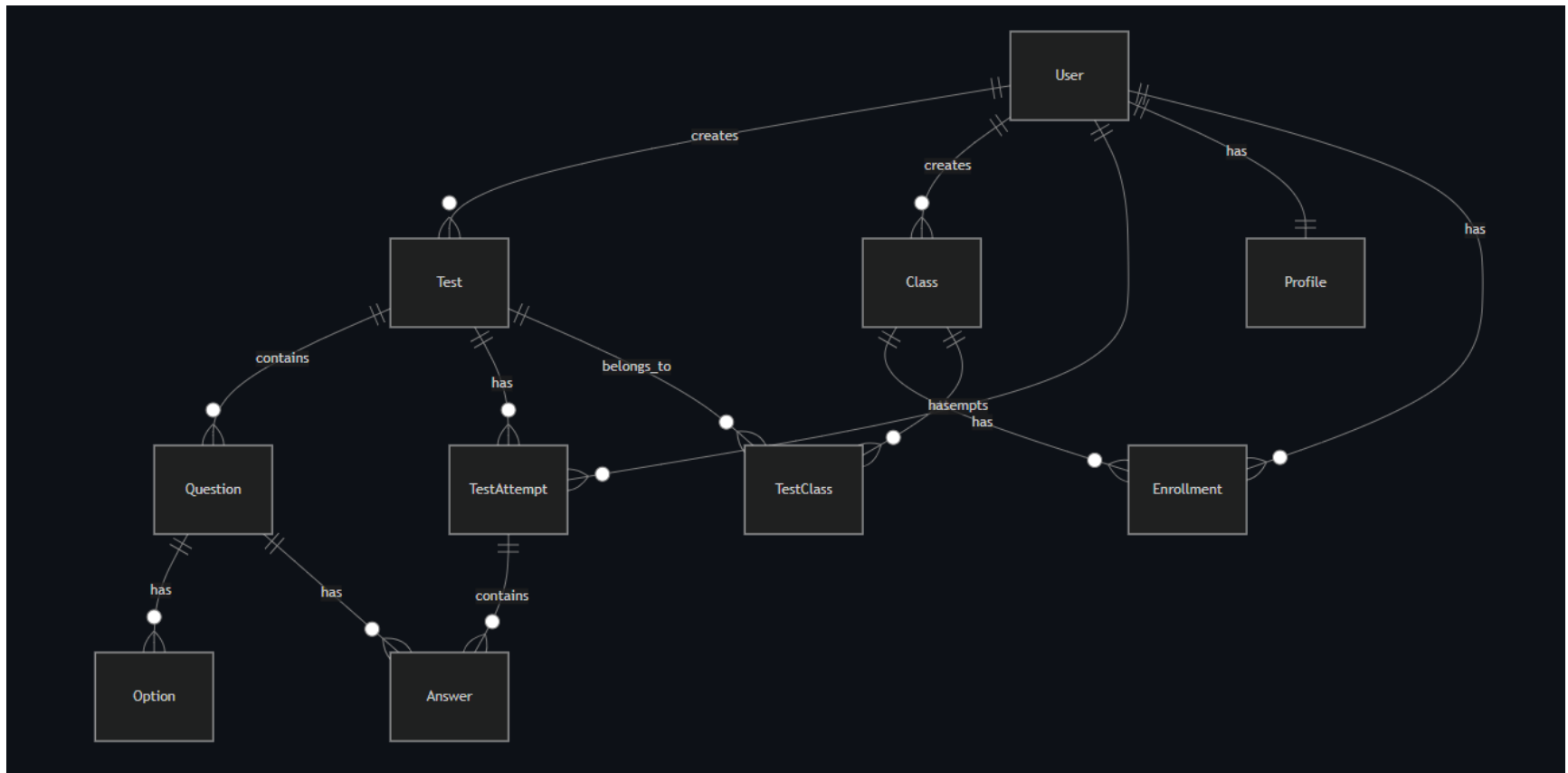
System Architecture Overview

React frontend with Express backend and PostgreSQL integration using JWT authentication and modular services.



Database Schema Overview

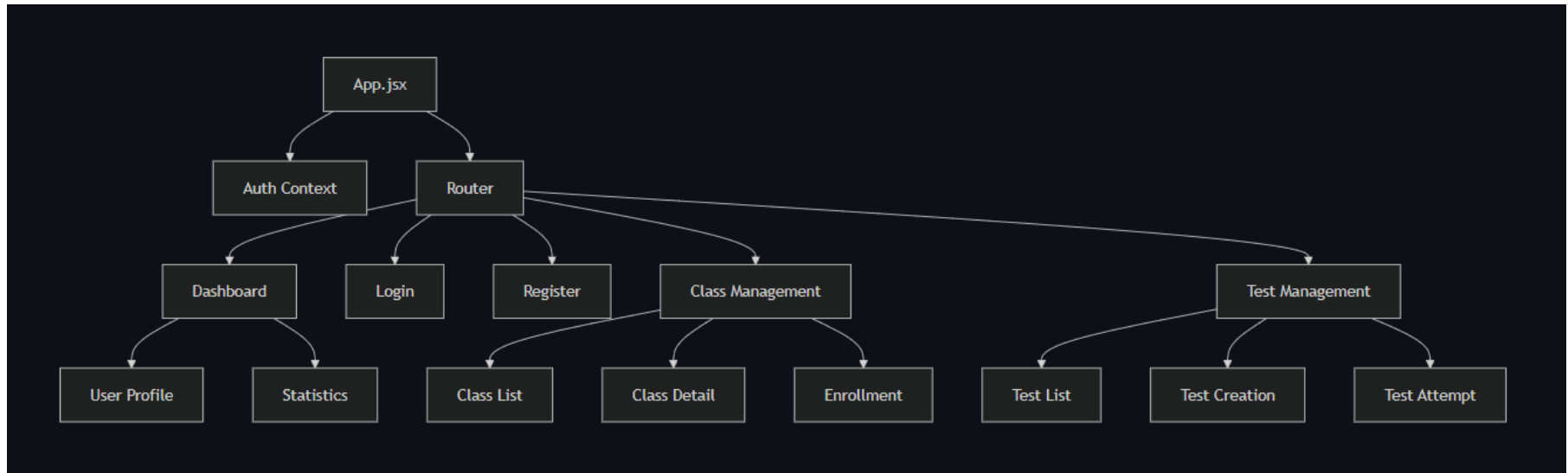
Entity-relationship diagram linking Users, Tests, Classes, and Attempts with relationships for questions, answers, enrollments, and profiles.





Component Structure Overview

Hierarchical layout of React components with App.jsx managing Auth Context and Routing for dashboard, class, and test management features.



RESULTS



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Most Popular

Practice Exams

Access hundreds of practice questions with detailed explanations

Instant Results

Get immediate feedback and performance analysis

Progress Tracking

Monitor your improvement with visual analytics

98%10K+
Pass RateStudents

Success Stories from Our Students 🌟

Join thousands of students who achieved their certification goals

92%

CCNA Exam

"I struggled with network protocols for months. This platform's practice tests identified my weak areas and helped me focus my study time efficiently. Passed on my first attempt!"

JD

James Davis
Network Engineer

98%

AWS Solutions Architect

"The analytics dashboard showed exactly where I needed to improve. After 4 weeks of targeted practice, I not only passed but scored in the top 5% nationally. Worth every penny!"

AR

Aisha Rahman
Cloud Solutions Architect

85%

CompTIA Security+

"Failed twice before finding this platform. The practice questions were remarkably similar to the actual exam, and the performance tracking helped me identify and eliminate my knowledge gaps."

MK

Mike Klein
IT Security Specialist

15,000+
Certified Students

94%
Pass Rate

250+
Exam Types

30%
Higher Scores

Home Page of Exam Matrix:

Clean and modern interface with easy navigation, a welcome message, and quick access to login or upcoming exams.

RESULTS

CHITKARA
UNIVERSITY



Exam Matrix

Sign in to your account

Email

Password

[Show](#)

☐ Remember me [Forgot Password?](#)

[Sign In](#)

Don't have an account? [Create account](#)

Need help? [Contact Support](#)

Create Your Account

Please fill in the details to get started

First Name

Last Name

Email Address

Password [Show](#)

Confirm Password

Select Your Role

☒ **Student**
Attempt exams and view results

☐ **Admin**
Manage exams, questions, and user records

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

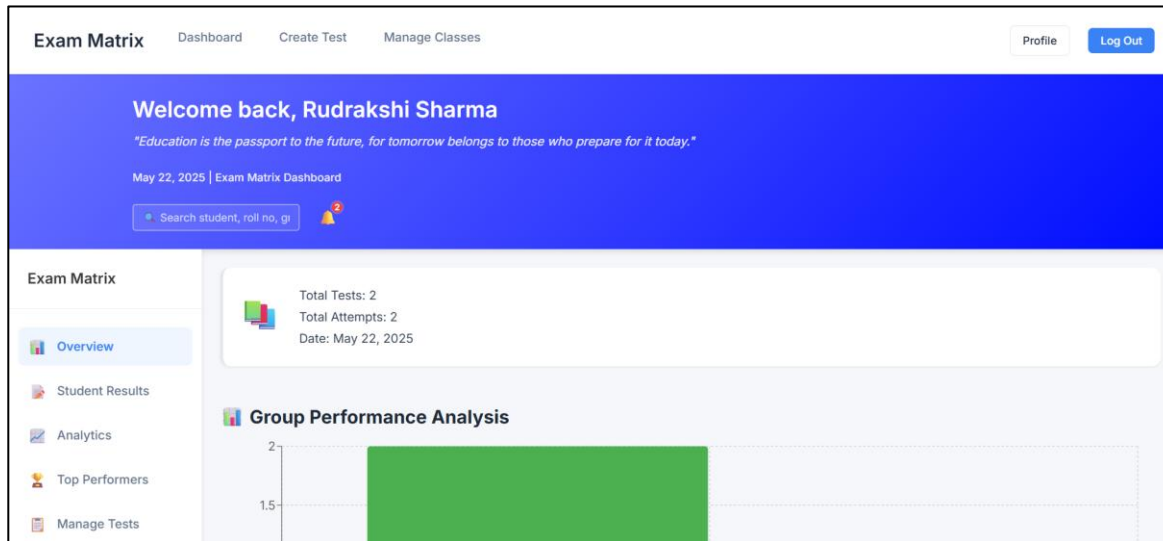
[Create Account](#)

Already have an account? [Sign In](#)

Login and Create Account Page:

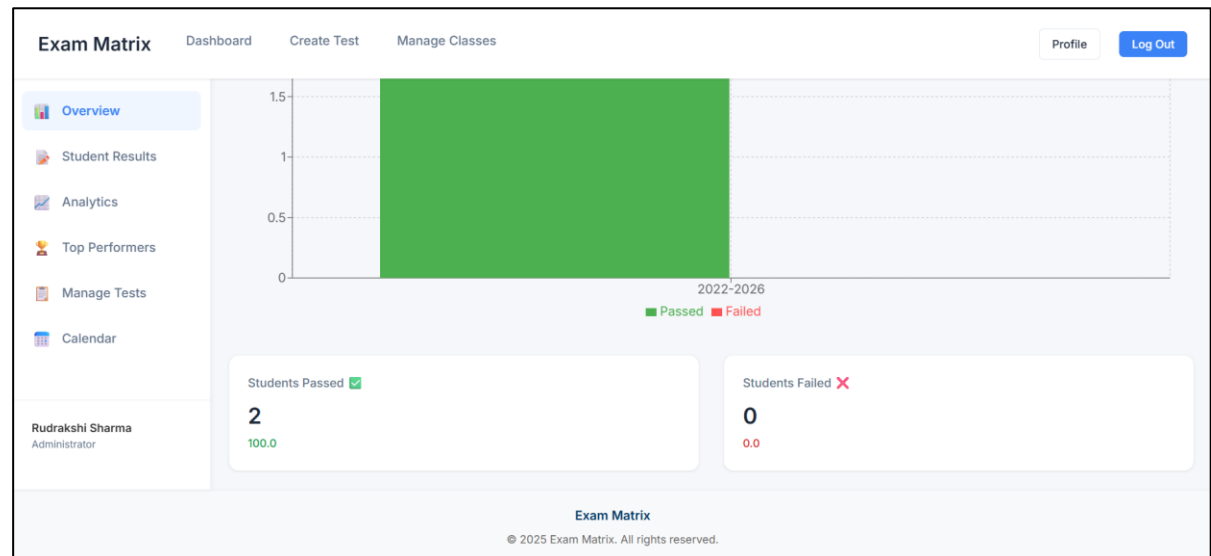
Allows users to sign in or register with role-based access for Admin and Student.

RESULTS



Admin Dashboard (Group Performance Analysis):

Shows detailed charts and metrics to track group exam results and progress.



RESULTS



Admin Dashboard

(Student Results):

Displays results of all students with sortable tables and summary statistics.

Exam Matrix

Overview

Student Results

Analytics

Top Performers

Manage Tests

Calendar

All Students Results

All Groups

All Grades

Sort by Marks

Sno.	Name	Roll Number	Group	Test	Marks	Grade	Status	Actions
1	Sakshi Rana	2210990769	2022-2026	22-May	100/100	A	PASSED	View Recording
2	Sakshi Rana	2210990769	2022-2026	20-May	67/100	D	PASSED	View Recording

Admin Dashboard (Advanced Analytics):

Provides in-depth data insights with interactive charts and trend analysis tools.

Exam Matrix

Overview

Student Results

Analytics

Top Performers

Manage Tests

Calendar

Advanced Analytics

Performance Summary

100.0% overall pass rate across all tests

Pass rate is above target of 70%

Test Statistics

2 total tests created

2 total test attempts

Student Engagement

1 unique students took tests

2 tests per student on average

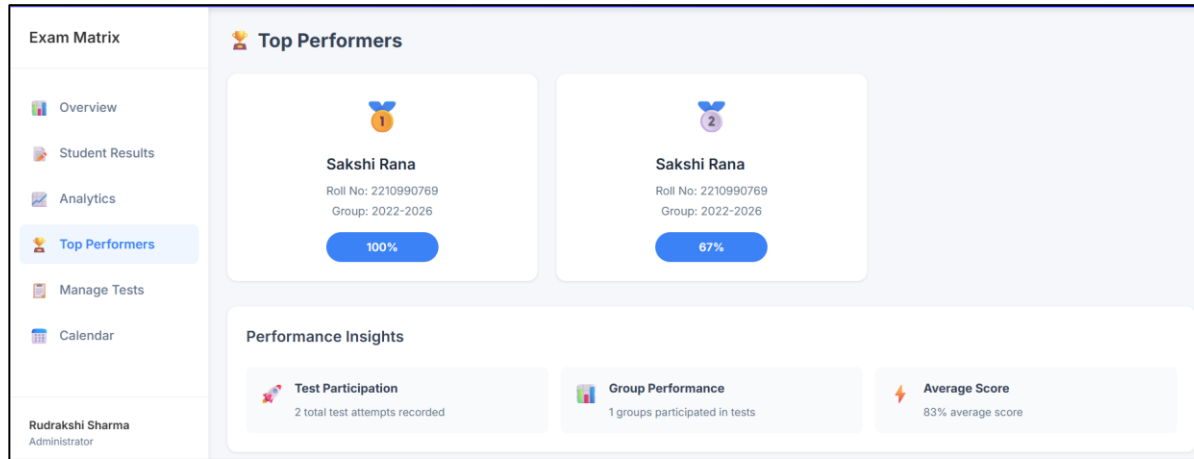
Score Distribution

90-100%		1 students (50%)
80-89%		0 students (0%)
70-79%		0 students (0%)
60-69%		1 students (50%)
50-59%		0 students (0%)
40-49%		0 students (0%)
Below 40%		0 students (0%)

Rudrakshi Sharma

Administrator

RESULTS



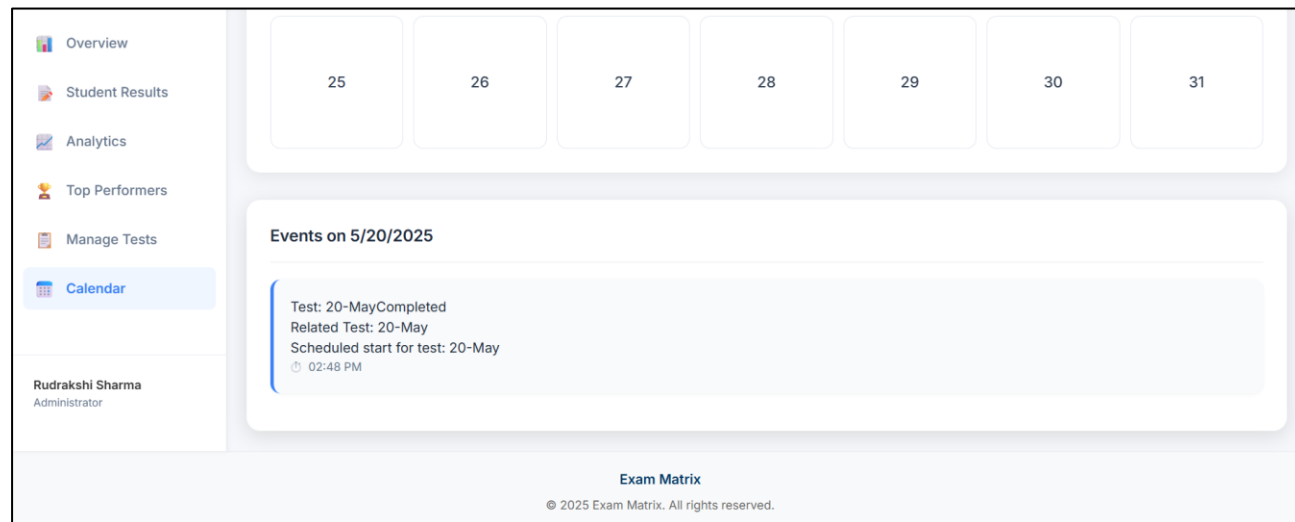
Admin Dashboard

(Top Performers):

Highlights highest-scoring students with rankings and performance summaries.

Calendar Interface (Admin):

Interactive scheduler for managing exams and important dates efficiently.



RESULTS



Create Test:

Enables admins to design exams by adding questions, setting time limits, and choosing formats.

The screenshot shows the 'Create Test' form in the Exam Matrix interface. The form is titled 'Create Test' and has a 'Save Test' button. It includes a 'Test Details' section with a 'Test Title' field (e.g., 'Midterm Exam - Data Structures') and a 'Test Description' field (Instructions or details about the test (optional)). Below this, there are fields for 'Duration (minutes) * *' (60) and 'Maximum Attempts' (1). There is a checkbox for 'Set start date/time' and a dropdown for 'Expiry' (Duration, Days, Never expires). A 'Require Recording' section at the bottom has a checkbox for 'Proctoring recording will be required for this test'.

This block provides a detailed view of the 'Create Test' form fields. It includes a 'Duration (minutes) * *' field with a value of 60, a 'Maximum Attempts' field with a value of 1, and a checkbox for 'Unlimited'. There is a checkbox for 'Set start date/time' and a dropdown for 'Expiry' (Duration, Days, Never expires). A 'Require Recording' section at the bottom has a checkbox for 'Proctoring recording will be required for this test'.

Test Settings:

Allows customization of test parameters like duration, access controls, and maximum attempts

RESULTS



Questions

Question 1

Multiple Choice ▾

☒ Required

Duplicate

Remove

Question Text *

What will be the output of the following C++ code?

```
#include <iostream>
using namespace std;
```

Options *

- ☐ 53 Remove
- ☒ 8 Remove
- ☐ a+3 Remove
- ☐ Compilation Error Remove

Add Option

Create Test MCQ Question:

Interface to add multiple-choice questions with options
and correct answer selection.

RESULTS



Question 2

Checkbox ▾☒ Required Duplicate Remove

Question Text *

Q2. Which of the following are valid variable names in C++?

Options *

☐ my-variable Remove

☒ _counter Remove

☐ 2ndValue Remove

☒ value_2 Remove

Add Option

Add Multiple Choice Question

Add Checkbox Question

Add Coding Question

Create Test Checkbox Question:

Interface to add multiple-answer questions where multiple options can be marked correct.

RESULTS

Test Preview (Admin View)

[Back to Editor](#)[Save Test](#)

Testing of 22 May

(No description)

Duration: 60 minutes

Attempts allowed: 1

Expires: Never

Available from: 5/22/2025, 6:45:00 PM

Question 1 * (MCQ)

What will be the output of the following C++ code? `#include <iostream> using namespace std; int main() { int a = 5; cout << a + 3; return 0; }`

- ☐ 53
- ☒ 8 ✓
- ☐ a+3
- ☐ Compilation Error

Question 2 * (CHECKBOX)

Q2. Which of the following are valid variable names in C++?

- ☐ my-variable
- ☒ _counter ✓
- ☐ 2ndValue
- ☒ value_2 ✓

Test Preview (Admin):

Allows admins to review the full exam layout and content before publishing.



Test Preview (Student View)

[Back to Editor](#)[Save Test](#)

Testing of 22 May

(No description)

Duration: 60 minutes

Attempts allowed: 1

Expires: Never

Available from: 5/22/2025, 6:45:00 PM

Question 1 * (MCQ)

What will be the output of the following C++ code? `#include <iostream> using namespace std; int main() { int a = 5; cout << a + 3; return 0; }`

Question 2 * (CHECKBOX)

Q2. Which of the following are valid variable names in C++?

Test Preview (Student):

Shows students a sample view of the exam format and question types before starting.

RESULTS



Exam Matrix

- Overview
- Student Results
- Analytics
- Top Performers
- Manage Tests**
- Calendar

Manage Tests

+ Create New Test

Testing of 22 May

60 mins 2 questions 1 attempts

No expiry Starts: 5/22/2025 **Inactive**

No description provided

Settings

22-May

60 mins 4 questions 1 attempts

No expiry Starts: 5/22/2025 **Complete**

No description provided

Settings

20-May

60 mins 3 questions 1 attempts

No expiry Starts: 5/20/2025 **Complete**

No description provided

Settings

Manage Test:

Displays a list of tests with options to activate, disable, publish, and configure recording settings.

Test Settings: Testing of 22 May

Test Status

Enable or disable this test for students

Disabled

Test Results

Release results to all students who attempted this test

Publish Results

Test Analytics

View detailed analytics for this test

View Analytics

Test Expiry

☒ No expiry

Set the number of days after which the test will expire

Update

☒ Schedule test start

Start Date & Time

Schedule when the test will become available

Date: 22-05-2025 Time: 18:45

Test Expiry

☒ No expiry

Set the number of days after which the test will expire

Update

☒ Schedule test start

Start Date & Time

Schedule when the test will become available

Date: 22-05-2025 Time: 18:45

☐ Unlimited attempts

Maximum Attempts

Set how many times a student can attempt the test.

1 attempts allowed

Update Attempts

☒ Recording Required

Require Recording

Specify if proctoring recording is mandatory for this test.

Update Recording Setting

Close

RESULTS



Class Management

Final Testing

No description provided

Class Code: 7BC226

0 Students 0 Tests



Create Class:

Allows admins to set up a new class with details and enrollment options.

Pending Requests:

Shows student join requests awaiting approval to enter the class.

← Manage Enrollments: Final Testing

ENROLLED STUDENTS **PENDING REQUESTS**

Student	Email	Roll Number	Class	Batch	Requested On	Actions
 Sakshi Rana	sakshi769.be22@chitkara.edu.in	2210990769	G-12	2022-2026	5/22/2025	 

Class Information

Class Code

7BC226

Share this code with students to allow them to join the class.

[← BACK TO CLASS DETAILS](#)

Enrolled Students:

Lists all students currently enrolled and active in the class.

← Manage Enrollments: Final Testing

ENROLLED STUDENTS PENDING REQUESTS

Student	Email	Roll Number	Class	Batch	Joined On
 Sakshi Rana	sakshi769.be22@chitkara.edu.in	2210990769	G-12	2022-2026	5/22/2025

Class Information

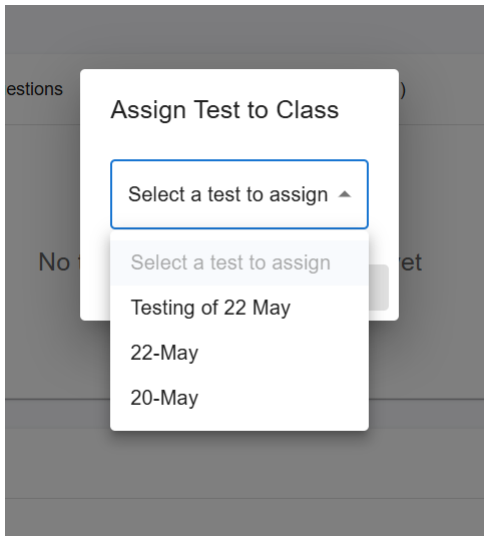
Class Code

7BC226

Share this code with students to allow them to join the class.

[← BACK TO CLASS DETAILS](#)

RESULTS



Assign Test:

Let Admins allocate exams to specific students or groups.

Admin Profile

Personal Information

First Name

Rudrakshi

Last Name

Sharma

Email

rudrakshisharma86@gmail.com

Password

Change Password

Save Changes

Admin Profile:

Displays admin's personal information and settings for account management.

RESULTS



My Classes

Final Testing

No description provided

Teacher: Rudrakshi Sharma

1 Tests

22-May

No description provided

Teacher: Rudrakshi Sharma

1 Tests

20-May

No description provided

Teacher: Rudrakshi Sharma

1 Tests

Join a Class

Enter the class code provided by your teacher to join a class.

JOIN CLASS

Class Status Guide

Approved

 You are enrolled in the class

Pending

 Your enrollment request is awaiting approval

Rejected

 Your enrollment request was rejected

My Classes (Student View):
Displays all the classes the student is currently enrolled in with quick access to details.

Assigned Classes (Student View):

Displays classes assigned to the student with details like Upcoming tests.

← Final Testing

Available Tests

Upcoming Tests

Testing of 22 May

Upcoming

No description provided

UPCOMING

60 minutes 2 questions

Scheduled: 5/22/2025 at 06:45 PM

Class Information

Description

No description provided

Teacher

Rudrakshi Sharma

Joined On

5/22/2025

Status

Enrolled

RESULTS



Test Calendar

< May 2025 >

25	26	27	28	29	30	31
----	----	----	----	----	----	----

Events & Tests This Month



Test: 20-May
5/20/2025 at 02:48 PM

 Completed

Scheduled start for test: 20-May



Testing of 22 May
5/22/2025 at 06:45 PM

 Scheduled

Duration: 60 minutes



Test: Testing of 22 May
5/22/2025 at 06:45 PM

 Scheduled

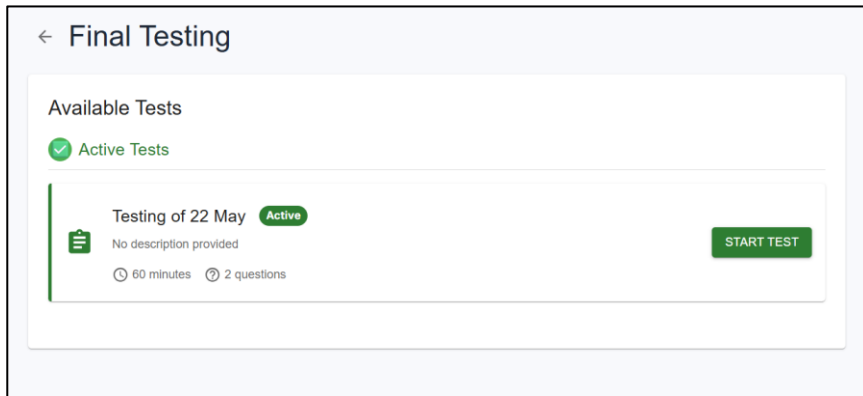
Scheduled start for test: Testing of 22 May

CLOSE

Calendar Interface (Student View):

Displays completed and upcoming tests marked on their respective dates for easy tracking.

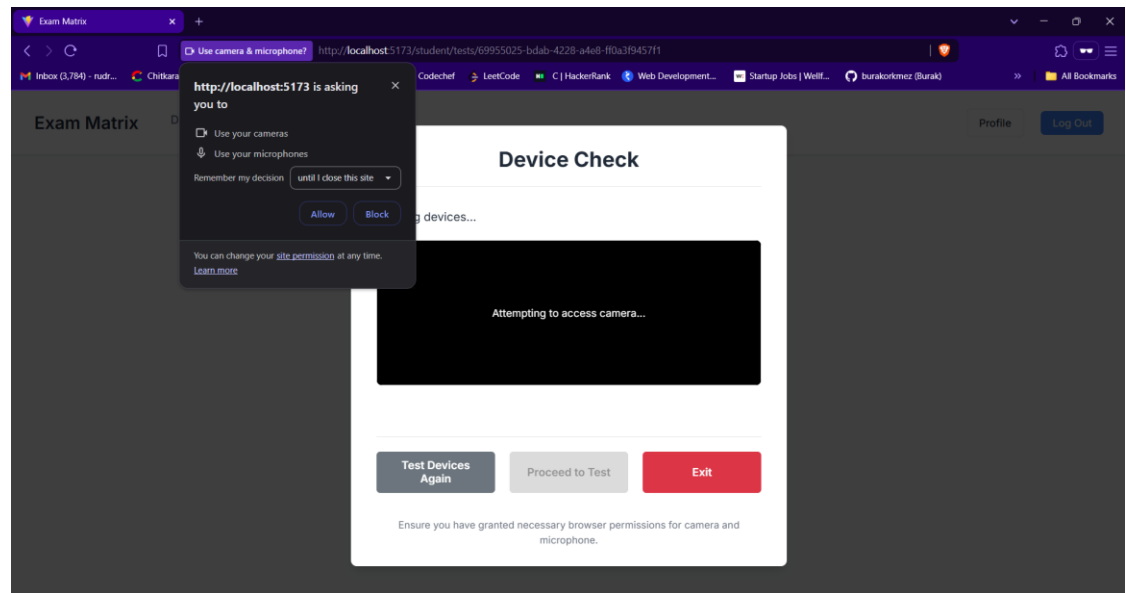
RESULTS



Available Tests (Student View):
Shows all tests currently accessible for the student to start or continue.

Camera and Microphone Permission:

Prompts the user to allow access to their camera and microphone for monitoring during the exam.



RESULTS

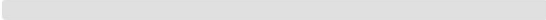


Camera:
USB2.0 HD UVC WebCam (3277:0029) ▼

Microphone:
Default - Headset (Airdopes 138) ▼



✓ Microphone Access Granted

Microphone Level:


Test Devices Again Proceed to Test Exit

Camera and Microphone Status:

Displays live indicators showing if the camera and microphone are active and recording.

Fullscreen Required:

Informs the user that the exam must be taken in full screen mode to prevent distractions or cheating.

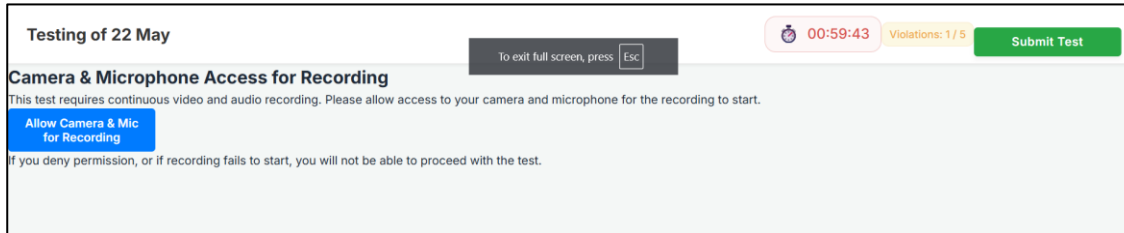
Fullscreen Required

This test must be taken in fullscreen mode to prevent distractions.

Enter Fullscreen

Exiting fullscreen during the test will count as a violation.

RESULTS

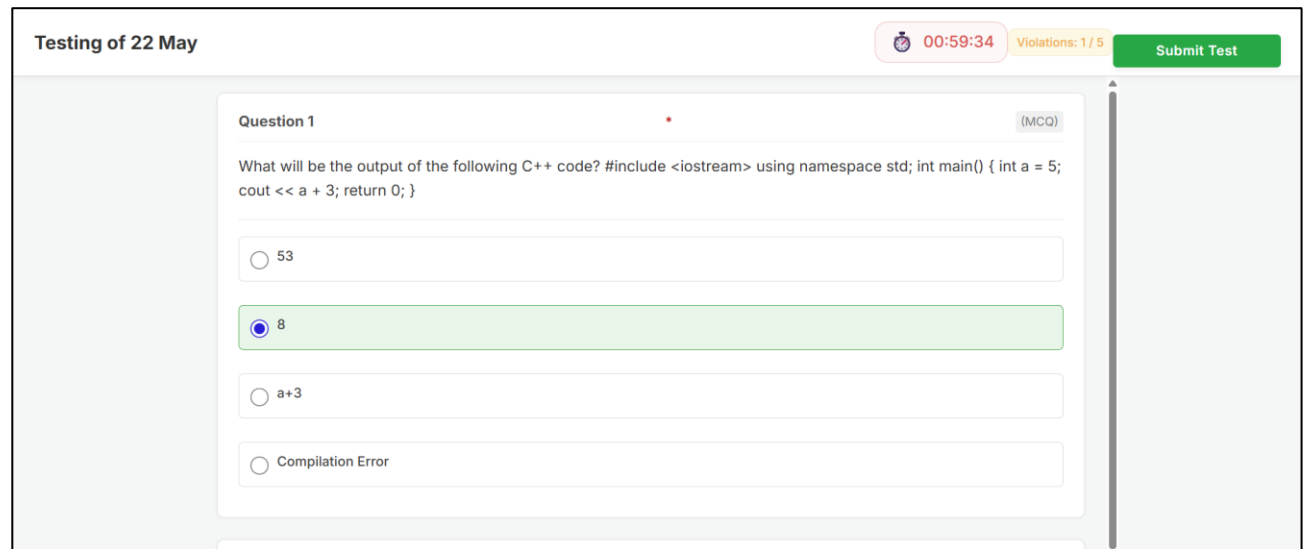


Requesting Camera and Microphone Permission:

Prompts the user again to grant access to their camera and microphone before starting the exam.

Attempt Test (MCQ):

Allows the student to select one correct option from multiple choices.



RESULTS



Question 2 (CHECKBOX)

Q2. Which of the following are valid variable names in C++?

☐ my-variable

☒ _counter

☒ 2ndValue

☐ value_2

Attempt Test (Checkbox):

Enables the student to select multiple correct answers from the given options.

Testing of 22 May

00:59:10 Violations: 3 / 5 Submit Test

Violation Detected (3/5): Switched Tab/Window

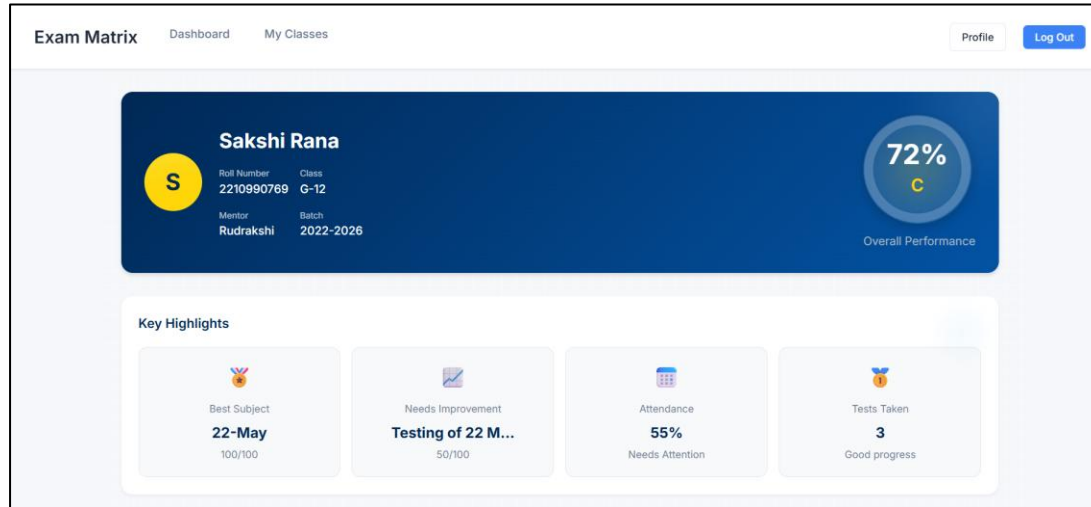
8

Violation Warning:

Alerts the student on actions like Ctrl+C, Ctrl+V, or tab switching — with a maximum of 5 violations allowed.

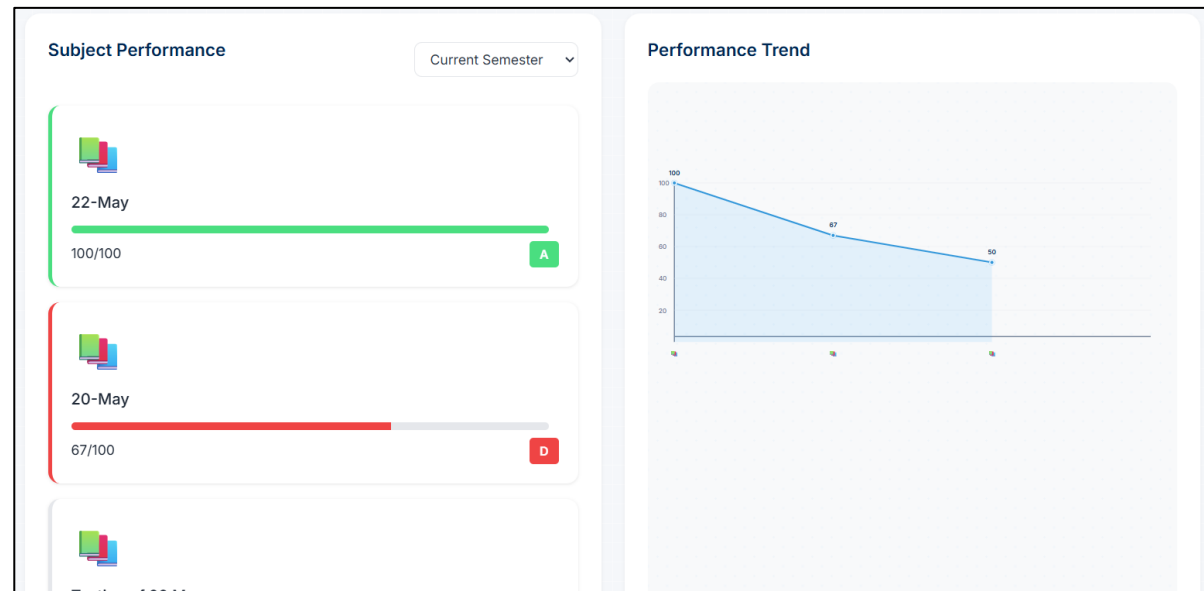
RESULTS

CHITKARA
UNIVERSITY



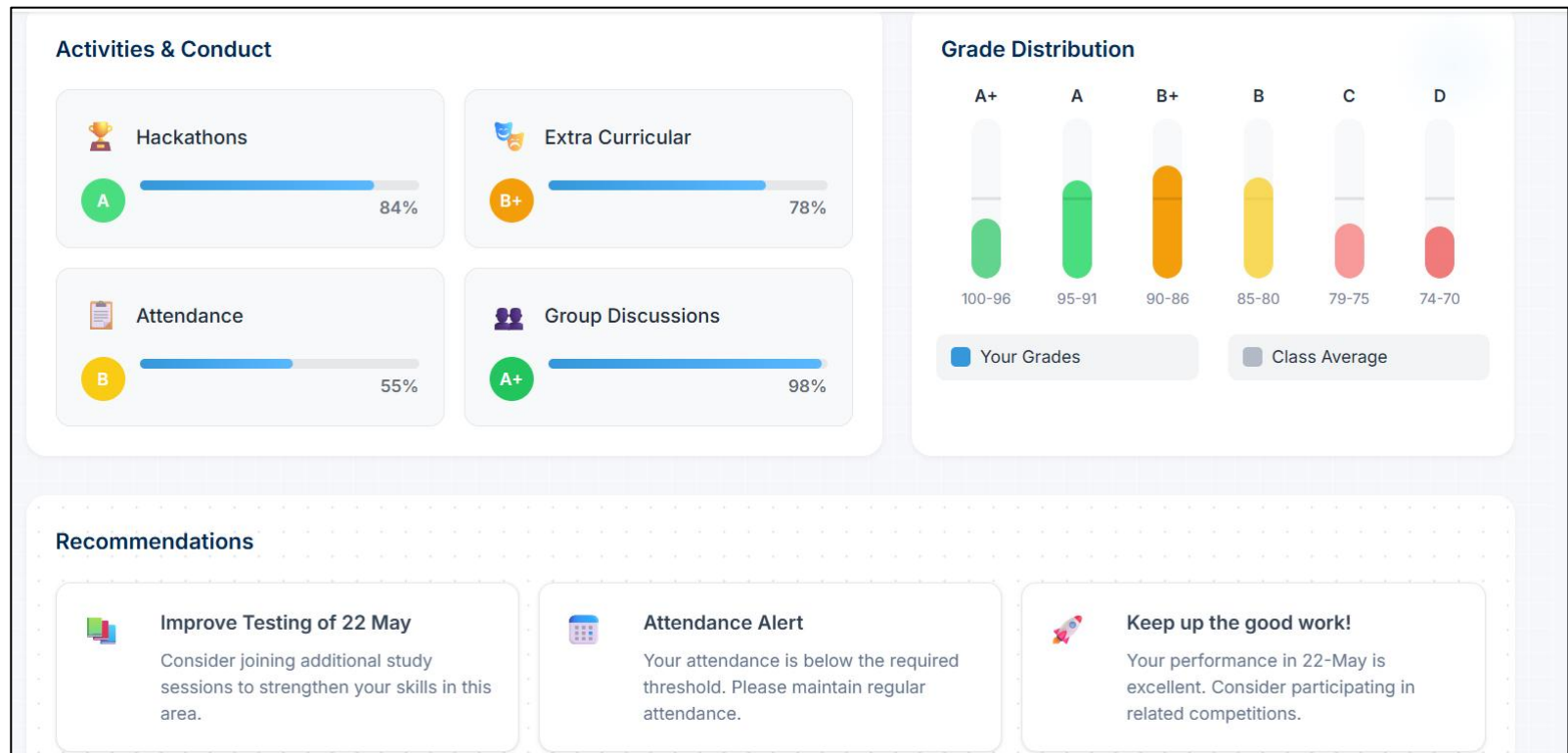
Student Dashboard:

Displays subject-wise results, highlights best and weak subjects, tracks test attempts, and gives a quick view of overall performance.



RESULTS

CHITKARA
UNIVERSITY



Student Dashboard:

Displays subject-wise results, highlights best and weak subjects, tracks test attempts, and gives a quick view of overall performance.

RESULTS

Student Profile

Personal Information

First Name

Sakshi

Last Name

Rana

Email

sakshi769.be22@chitkara.edu.in

Academic Information

Roll Number

2210990769

Class/Grade

G-12

Batch/Year

2022-2026

Mentor

Rudrakshi

Password

[Change Password](#)

[Save Changes](#)

Student Profile:

Shows personal details , like Mentor , Batch details and allows updating account information and preferences.

FUTURE SCOPE

- **Multi-language Coding & AI Feedback:**

Enable students to write and test code in various programming languages like Python, Java, C++, and more. Integrate AI to provide real-time hints, detect common mistakes, and offer suggestions to improve code quality and efficiency.

- **AI Proctoring:**

Implement advanced AI-driven proctoring that uses facial recognition, eye movement tracking, and behavior analysis to monitor exam integrity in real-time, flagging suspicious activities automatically to minimize cheating.

- **Gamification:**

Introduce badges, leaderboards, and milestone rewards to create a fun and competitive learning environment. Students can earn points for completing tests, improving scores, and participating in challenges, motivating continuous engagement and improvement.

- **Mobile App:**

Develop a user-friendly mobile application that allows students to take exams, check results, and receive notifications on the go. This makes the platform accessible anytime and anywhere, increasing flexibility and convenience for learners.

CONCLUSION

Exam Matrix represents a significant advancement in online examination systems, offering a comprehensive solution that addresses the limitations of traditional assessment methods. By providing a secure, flexible, and analytical platform for conducting examinations, we've created a tool that benefits both educational institutions and students alike.

The system's role-based access, multi-format question support, robust security measures, and detailed analytics dashboard represent a complete transformation of the examination process. Our future enhancements, including expanded question formats, advanced security features, and additional functionality like calendar integration, will further solidify Exam Matrix as a leading solution in educational assessment.

As we continue to develop and refine this platform, we remain committed to enhancing the examination experience through innovation, security, and actionable insights. Thank you for your attention, and we welcome any questions or feedback you may have.

REFERENCES

- 1) React.js Documentation – A JavaScript library for building responsive user interfaces.
<https://reactjs.org/>
- 2) CSS Documentation – Stylesheet language for designing web pages.
<https://www.w3schools.com/css/>
- 3) Tanstack React Query Documentation – Library for API fetching and caching in React.
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- 6) Prisma ORM Documentation – Type-safe database query tool for Node.js.
<https://www.prisma.io/>
- 7) PostgreSQL Documentation – Open-source relational database system.
<https://www.postgresql.org/>
- 8) JWT Documentation – Compact token for secure authentication.
<https://jwt.io/>
- 9) Neon DB Documentation – Cloud-native serverless PostgreSQL database.
<https://neon.tech/>



***THANK
YOU***